

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
K. D. PATEL DEPARTMENT OF INFORMATION TECHNOLOGY
ACADEMIC YEAR: 2019-20

List of Experiments

Name of Teacher:Ravi Patel

Date: 12-07-2019

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Subject Name: Advanced Computing

Semester: 7

Subject Code: IT442

Academic Year: 2019-20

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Sr. No.	Practical	Hrs
1.	Setup High Performance MPI Cluster using MPICH library and running MPI Application Program. (minimum two nodes) Reference: https://sandip9334.blogspot.com/2016/07/mpich-multi-node-cluster-configuration.html	
2.	Creating and Accessing Remote Procedure Call (Service) using gRPC Framework. (Scalable Micro service). Reference: https://scotch.io/tutorials/implementing-remote-procedure-calls-with-grpc-and-protocol-buffers	02
3.	Set Up Hadoop Cluster and run news-paper word count program using Map-Reduce. Reference: https://www.michael-noll.com/tutorials/running-hadoop-on-ubuntu-linux-multi-node-cluster/	04
4.	Creating and using Virtual Machine using Xen Hypervisor (Private Cloud) Reference: http://snpatel.in/2018/12/29/xen-server-7-2-0-installation-steps/	04
5.	Implementing Load balancing/High Availability for Web Application using Amazon Web Services (AWS) Reference:	04
6.	Web-application deployment on EC2 using AWS CodeDeploy & S3 services. (with CI/CD) Reference: http://snpatel.in/2019/03/24/php-application-deployment-on-ec2-using-aws-codedeploy-s3/	02
7.	Implement Docker Containers. Reference: https://docs.docker.com/get-started/	04
8.	Implement Kubernetes Cluster. Reference: https://www.youtube.com/watch?v=6xJwQgDnMFE	02

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9.	Create your “ownCloud” and perform security testing on it. Reference: https://doc.owncloud.org/server/10.2/admin_manual/installation/	02
10.	Case Study and Presentation of Framework/Tool/Project.	04